Express and HTTP with real world examples
Postman

Going to the doctor

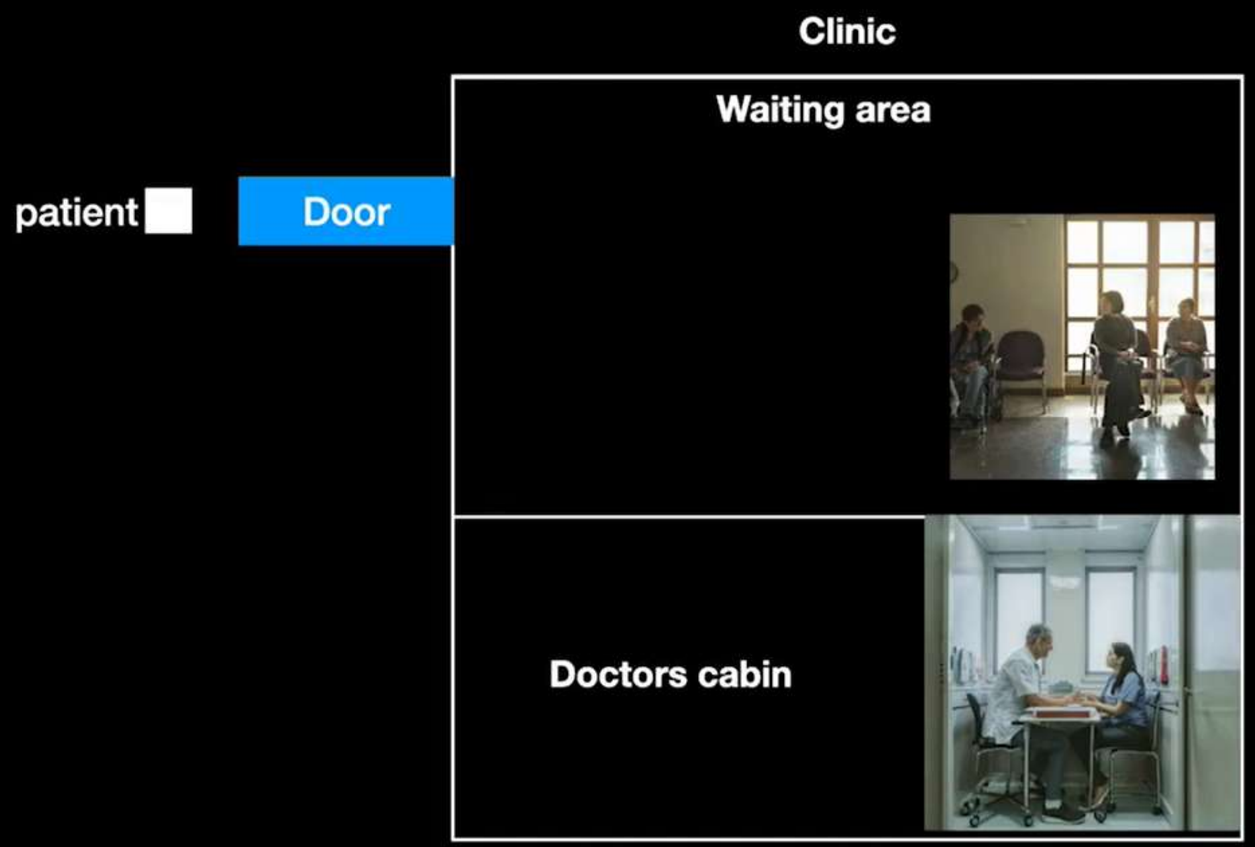
Doctors have a skill
They have acquired that skill over years
They provide service to other people who want to use their skill

Going to the doctor

To expose this life skill, they open a clinic
People who want to use their skill line up in a waiting room
One by one, the doctor meets with them
The doctor is single threaded

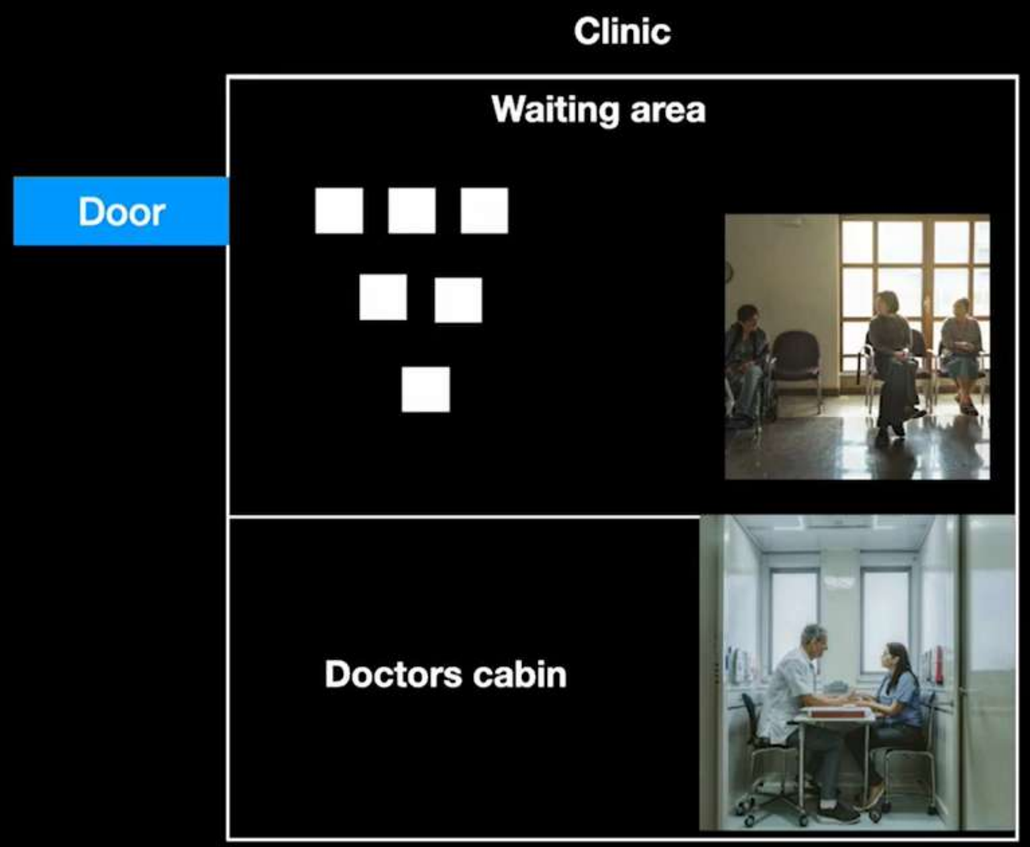
Going to the doctor

How do people reach the doctors?
They get their address and navigate to it



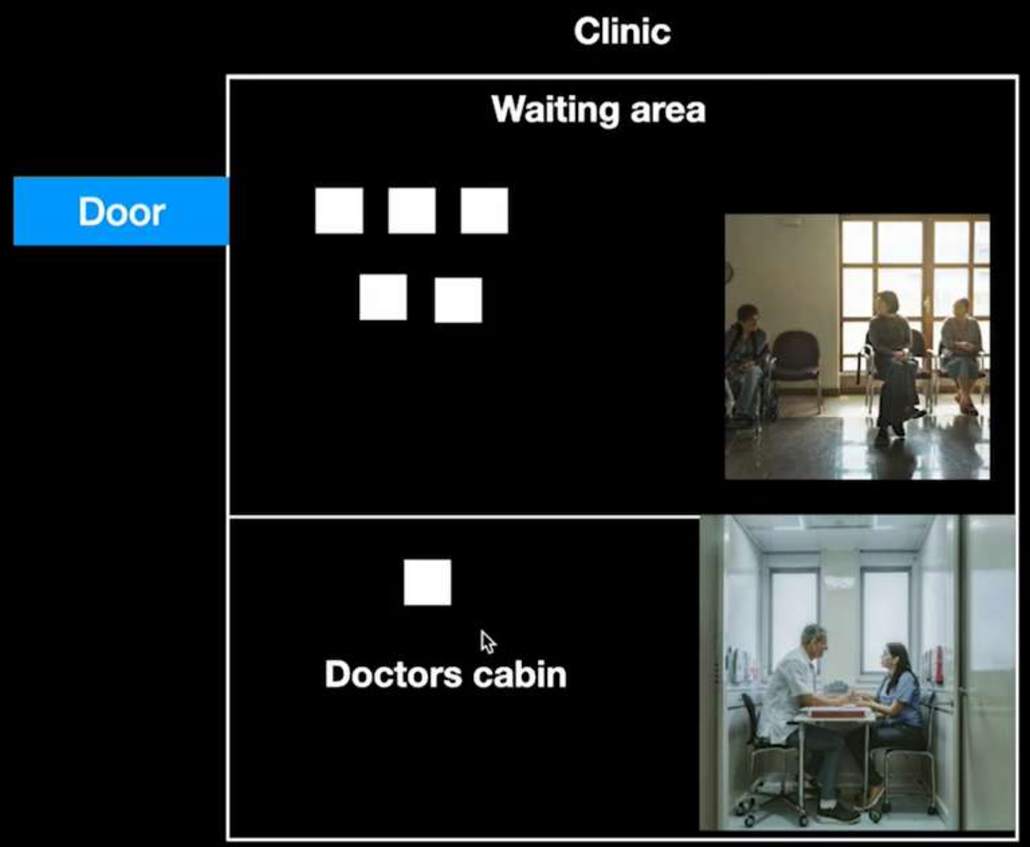
Going to the doctor

Once they reach there, they wait in the waiting area
Until their time comes



Going to the doctor

Doctor tends to them one by one



Going to the doctor

Clinic

Waiting area

Door

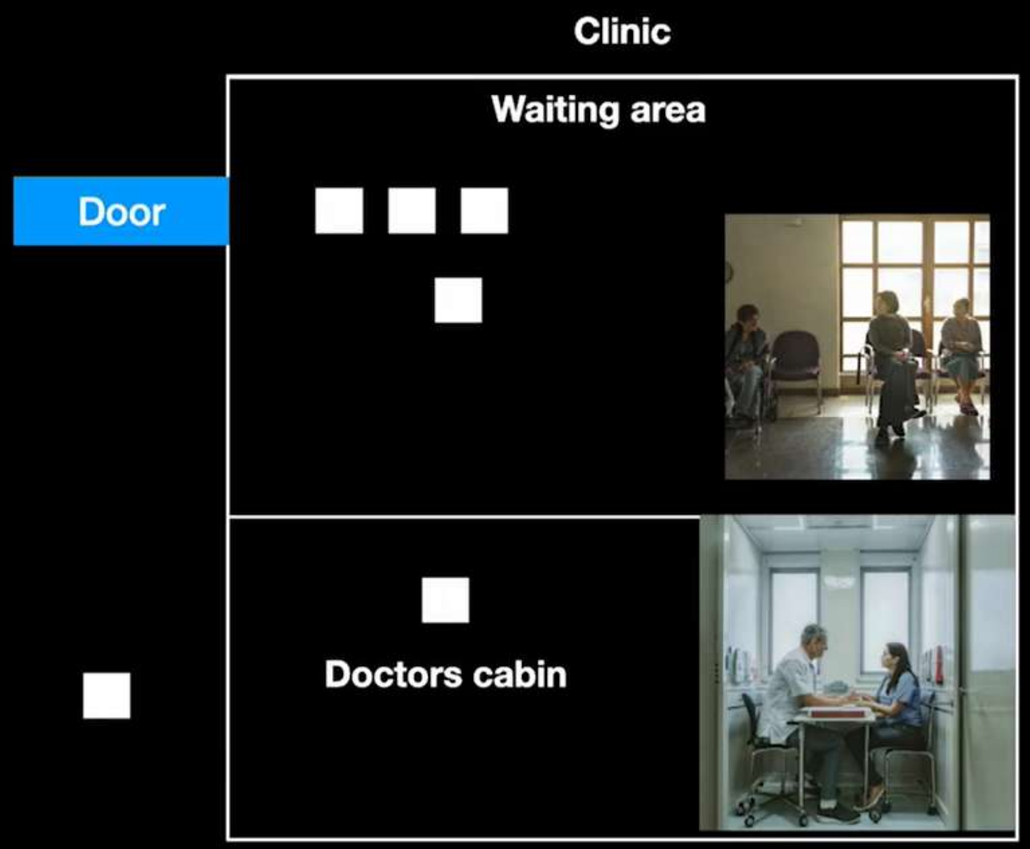
Doctor can tell them to get a medicine in the middle and meanwhile tend to other people

Doctors cabin

Go get medicine from chemist

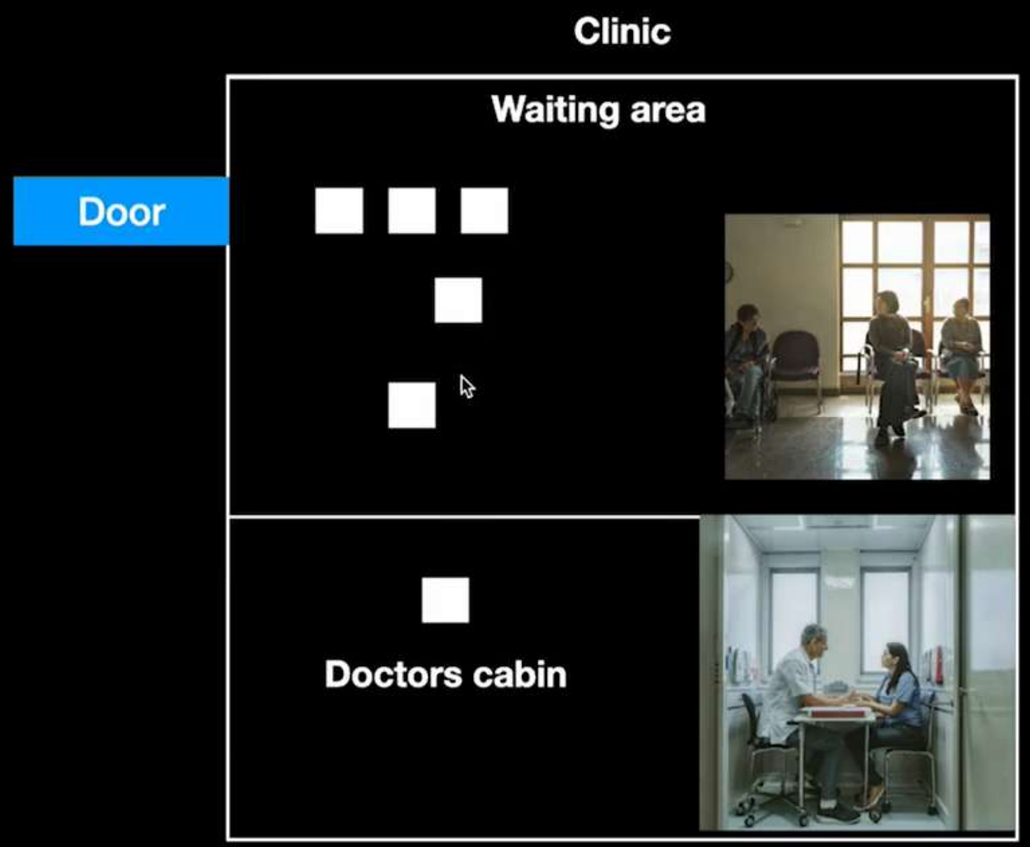
Going to the doctor

You Come back and wait in the waiting room again



Going to the doctor

You Come back and wait in the waiting room again



Your logic is like a doctor

Doctor logic

index.js > ...

```js
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

Your relative using you like a patient
Relative doesn't need to find your address,
They stay in the same house

Your logic is like a doctor

But what if you want to expose this logic to the world?

Doctor logic

index.js > ...

```javascript
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

Your logic is like a doctor

But what if you want to expose this logic to the world?
This is where HTTP comes into the picture
It lets you create a ~hostpital where people can
Come and find you

Doctor logic

index.js > ...

```javascript
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

Your logic is like a doctor

Question - How do I expose my doctor functionality
To other people?
How can they find me?

Ans - By creating an HTTP Server

Doctor logic

index.js > ...

```js
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

Doctor logic

index.js > ...

```javascript
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

index.js > ...

```javascript
function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

let ans = calculateSum(10);
console.log(ans);
```

index.js

```javascript
const express = require("express")

function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

const app = express();

app.get("/", function(req, res) {
  const n = req.query.n;
  const ans = calculateSum(n)
  res.send(ans);
})

app.listen(3000);
```

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

```js
const express = require("express")

function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

const app = express();

app.get("/", function(req, res) {
  const n = req.query.n;
  const ans = calculateSum(n)
  res.send(ans);
})

app.listen(3000);
```

Exposing the doctors one functionality (kidney surgery, brain surgery)
Doctor could have multiple rooms inside their hospital, this is one of them

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

```
const express = require("express")

function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

const app = express();

app.get("/", function(req, res) {
  const n = req.query.n;
  const ans = calculateSum(n)
  res.send(ans);
})

app.listen(3000);
```

If you take doctor's example the doctor can do many surgeries related to heart, kidney etc here for example the doctor is doing heart surgery in the room number 3000. Another kidney surgery is doing in room 3001

Deciding the address of the clinic

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

Hospital

Doctor 1

index.js > ...

```javascript
const express = require("express")

function calculateSum(a, b) {
  return a + b;
}

const app = express();

app.get("/", function(req, res) {
  const a = req.query.a;
  const b = req.query.b;
  const ans = calculateSum(a, b)
  res.send(ans);
})

app.listen(3001);
```

Doctor 2

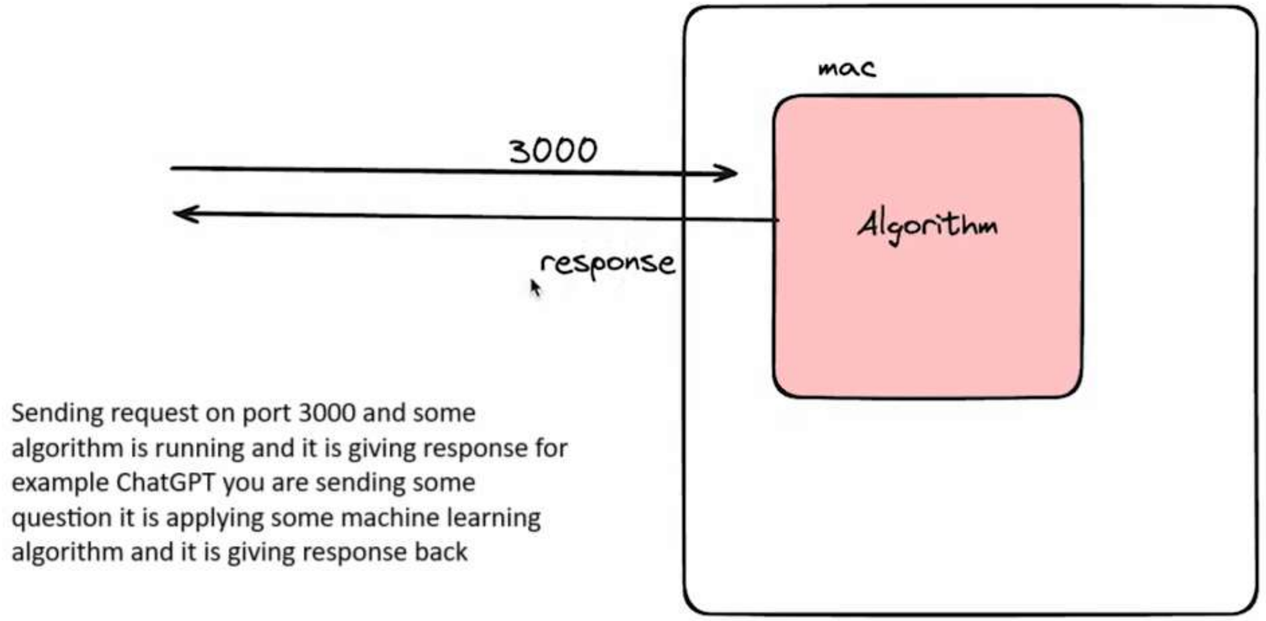
```javascript
const express = require("express")

function calculateSum(n) {
  let ans = 0;
  for (let i = 1; i<=n; i++) {
    ans = ans + i;
  }
  return ans;
}

const app = express();

app.get("/", function(req, res) {
  const n = req.query.n;
  const ans = calculateSum(n)
  res.send(ans);
})

app.listen(3000);
```



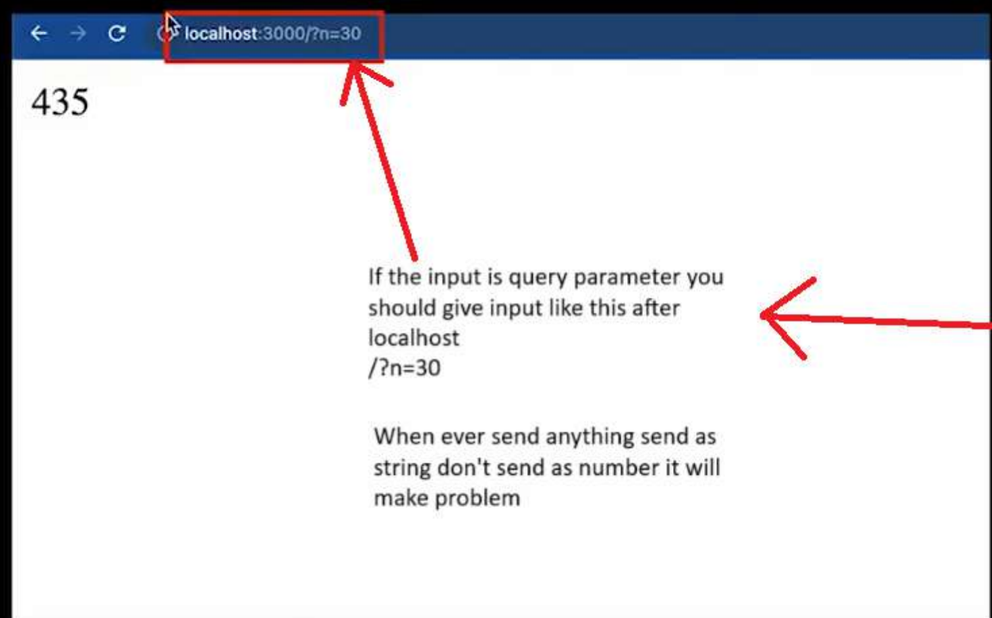
Sending request on port 3000 and some algorithm is running and it is giving response for example ChatGPT you are sending some question it is applying some machine learning algorithm and it is giving response back

Your logic is like a doctor

Question - How do I create an HTTP Server?

Ans - Express

How do patients reach it?



index.js > ...

```javascript
const express = require("express")

function calculateSum(n) {
  let ans = 0;
  for (let i = 0; i<n; i++) {
    ans = ans + i;
  }
  return ans;
}

const app = express();

app.get("/", function(req, res) {
  const n = req.query.n;
  const ans = calculateSum(n)
  res.send(ans.toString());
}

app.listen(3000);
```

OUTPUT DEBUG CONSOLE TERMINAL 1

TERMINAL

```
http-server-2 node index.js
```

Your logic is like a doctor

Request methods

1. **GET - Going for a consultation to get a check up**
2. **POST - Going to get a new kidney inserted**
3. **PUT - Going to get a kidney replaced**
4. **DELETE - Going to get a kidney removed**

Your logic is like a doctor

Status codes

1. 200 - Everything went fine
2. 404 - Doctor is not in the hospital
3. 500 - Mid surgery light went away
4. 411 - Inputs were incorrect, wrong person came to surgery
5. 403 => you were not allowed in the hospital

Your logic is like a doctor

Learn by doing, lets create an in memory hospital

You need to create 4 routes (4 things that the hospital can do)

1. GET - User can check how many kidneys they have and their health
2. POST - User can add a new kidney
3. PUT - User can replace a kidney, make it healthy
4. DELETE - User can remove a kidney

Your logic is like a doctor

Learn by doing, lets create an in memory hospital

Lets start by creating an in memory array that looks something like this -

```
1  var users = [{
2    name: 'John',
3    kidneys: [{
4      healthy: false
5    }, {
6      healthy: true
7    }]
8  }]
9
10 console.log(users[0]);
```

Your logic is like a doctor

Learn by doing, lets create an in memory hospital

You need to create 4 routes (4 things that the hospital can do)

1. **GET - User can check how many kidneys they have and their health**
2. **POST - User can add a new kidney**
3. **PUT - User can replace a kidney, make it healthy**
4. **DELETE - User can remove a kidney**

index.js > ...

```javascript
const express = require("express")
const app = express();
var users = [{
    name: 'John',
    kidneys: [{
        healthy: false
    }, {
        healthy: true
    }]
}]

app.get("/", function(req, res) {
                ?
})

app.post("/", function(req, res) {

})

app.put("/", function(req, res) {

})

app.delete("/", function(req, res) {

})

app.listen(3000);
```

Learn by doing, lets create an in memory hospital

1. What should happen if they try to delete when there are no kidneys?
2. What should happen if they try to make a kidney healthy when all are already healthy